

Question ID 0d685865

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	

ID: 0d685865

SAT4 M 1.1

If $x = 7$, what is the value of $x + 20$?

- A. 13
- B. 20
- C. 27
- D. 34

ID: 0d685865 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that $x = 7$. Substituting 7 for x into the given expression $x + 20$ yields $7 + 20$, which is equivalent to 27.

Choice A is incorrect. This is the value of $x + 6$.

Choice B is incorrect. This is the value of $x + 13$.

Choice D is incorrect. This is the value of $x + 27$.

Question Difficulty: Easy

Question ID 55cfaf22

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 55cfaf22

SAT4 M 1.2

Data set X: 5, 9, 9, 13

Data set Y: 5, 9, 9, 13, 27

The lists give the values in data sets X and Y. Which statement correctly compares the mean of data set X and the mean of data set Y?

- A. The mean of data set X is greater than the mean of data set Y.
- B. The mean of data set X is less than the mean of data set Y.
- C. The means of data set X and data set Y are equal.
- D. There is not enough information to compare the means.

ID: 55cfaf22 Answer

Correct Answer: B

Rationale

Choice B is correct. The mean of a data set is the sum of the values in the data set divided by the number of values in the data set. It follows that the mean of data set X is $\frac{5+9+9+13}{4}$, or 9, and the mean of data set Y is $\frac{5+9+9+13+27}{5}$, or 12.6. Since 9 is less than 12.6, the mean of data set X is less than the mean of data set Y.

Alternate approach: Data set Y consists of the 4 values in data set X and one additional value, 27. Since the additional value, 27, is larger than any value in data set X, the mean of data set X is less than the mean of data set Y.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

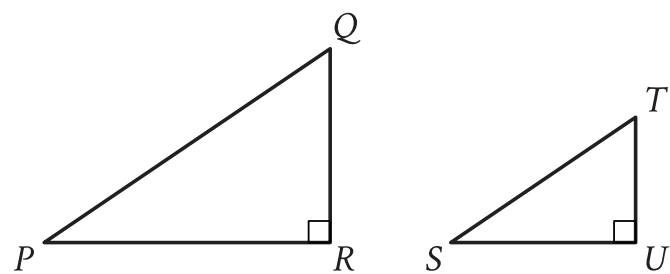
Question Difficulty: Easy

Question ID d5f349b7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: d5f349b7

SAT4 M 1.3



Note: Figures not drawn to scale.

Right triangles PQR and STU are similar, where P corresponds to S . If the measure of angle Q is 18° , what is the measure of angle S ?

- A. 18°
- B. 72°
- C. 82°
- D. 162°

ID: d5f349b7 Answer

Correct Answer: B

Rationale

Choice B is correct. In similar triangles, corresponding angles are congruent. It's given that right triangles PQR and STU are similar, where angle P corresponds to angle S . It follows that angle P is congruent to angle S . In the triangles shown, angle R and angle U are both marked as right angles, so angle R and angle U are corresponding angles. It follows that angle Q and angle T are corresponding angles, and thus, angle Q is congruent to angle T . It's given that the measure of angle Q is 18° , so the measure of angle T is also 18° . Angle U is a right angle, so the measure of angle U is 90° . The sum of the measures of the interior angles of a triangle is 180° . Thus, the sum of the measures of the interior angles of triangle STU is 180 degrees. Let s represent the measure, in degrees, of angle S . It follows that $s + 18 + 90 = 180$, or $s + 108 = 180$. Subtracting 108 from both sides of this equation yields $s = 72$. Therefore, if the measure of angle Q is 18 degrees, then the measure of angle S is 72 degrees.

Choice A is incorrect. This is the measure of angle T .

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the sum of the measures of angle S and angle U .

Question Difficulty: Easy

Question ID fbb0ea7f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	

ID: fbb0ea7f

SAT4 M 1.4

A rocket contained **467,000** kilograms (kg) of propellant before launch. Exactly **21** seconds after launch, **362,105** kg of this propellant remained. On average, approximately how much propellant, in kg, did the rocket burn each second after launch?

- A. **4,995**
- B. **17,243**
- C. **39,481**
- D. **104,895**

ID: fbb0ea7f Answer

Correct Answer: A

Rationale

Choice A is correct. It’s given that the rocket contained **467,000 kilograms (kg)** of propellant before launch and had **362,105 kg** remaining exactly **21** seconds after launch. Finding the difference between the amount, in **kg**, of propellant before launch and the remaining amount, in **kg**, of propellant after launch gives the amount, in **kg**, of propellant burned during the **21** seconds: $467,000 - 362,105 = 104,895$. Dividing the amount of propellant burned by the number of seconds yields $\frac{104,895}{21} = 4,995$. Thus, an average of **4,995 kg** of propellant burned each second after launch.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from finding the amount of propellant burned, rather than the amount of propellant burned each second.

Question Difficulty: Easy

Question ID 9db5b5c1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 9db5b5c1

SAT4 M 1.5

$$\begin{aligned}4x &= 20 \\ -3x + y &= -7\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of $x + y$?

- A. -27
- B. -13
- C. 13
- D. 27

ID: 9db5b5c1 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that $4x = 20$ and $-3x + y = -7$ is a system of equations with a solution (x, y) . Adding the second equation in the given system to the first equation yields $4x + (-3x + y) = 20 + (-7)$, which is equivalent to $x + y = 13$. Thus, the value of $x + y$ is 13 .

Choice A is incorrect. This represents the value of $-2(x + y) - 1$.

Choice B is incorrect. This represents the value of $-(x + y)$.

Choice D is incorrect. This represents the value of $2(x + y) + 1$.

Question Difficulty: Easy

Question ID 686b7244

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	■ ■ ■

ID: 686b7244

SAT4 M 1.6

A certain apprentice has enrolled in **85** hours of training courses. The equation **$10x + 15y = 85$** represents this situation, where **x** is the number of on-site training courses and **y** is the number of online training courses this apprentice has enrolled in. How many more hours does each online training course take than each on-site training course?

ID: 686b7244 Answer

Correct Answer: 5

Rationale

The correct answer is **5**. It's given that the equation **$10x + 15y = 85$** represents the situation, where **x** is the number of on-site training courses, **y** is the number of online training courses, and **85** is the total number of hours of training courses the apprentice has enrolled in. Therefore, **$10x$** represents the number of hours the apprentice has enrolled in on-site training courses, and **$15y$** represents the number of hours the apprentice has enrolled in online training courses. Since **x** is the number of on-site training courses and **y** is the number of online training courses the apprentice has enrolled in, **10** is the number of hours each on-site course takes and **15** is the number of hours each online course takes. Subtracting these numbers gives **$15 - 10$** , or **5** more hours each online training course takes than each on-site training course.

Question Difficulty: Hard

Question ID 1f0b582e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 1f0b582e

SAT4 M 1.7

Square X has a side length of **12** centimeters. The perimeter of square Y is **2** times the perimeter of square X. What is the length, in centimeters, of one side of square Y?

- A. **6**
- B. **10**
- C. **14**
- D. **24**

ID: 1f0b582e Answer

Correct Answer: D

Rationale

Choice D is correct. The perimeter, P , of a square can be found using the formula $P = 4s$, where s is the length of each side of the square. It's given that square X has a side length of **12** centimeters. Substituting **12** for s in the formula for the perimeter of a square yields $P = 4(12)$, or $P = 48$. Therefore, the perimeter of square X is **48** centimeters. It's also given that the perimeter of square Y is **2** times the perimeter of square X. Therefore, the perimeter of square Y is **2(48)**, or **96**, centimeters. Substituting **96** for P in the formula $P = 4s$ gives $96 = 4s$. Dividing both sides of this equation by **4** gives **24** = s . Therefore, the length of one side of square Y is **24** centimeters.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 1bc11c4e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 1bc11c4e

SAT4 M 1.8

$$g(m) = -0.05m + 12.1$$

The given function g models the number of gallons of gasoline that remains from a full gas tank in a car after driving m miles. According to the model, about how many gallons of gasoline are used to drive each mile?

- A. 0.05
- B. 12.1
- C. 20
- D. 242.0

ID: 1bc11c4e Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the function g models the number of gallons that remain from a full gas tank in a car after driving m miles. In the given function $g(m) = -0.05m + 12.1$, the coefficient of m is -0.05 . This means that for every increase in the value of m by 1, the value of $g(m)$ decreases by 0.05. It follows that for each mile driven, there is a decrease of 0.05 gallons of gasoline. Therefore, 0.05 gallons of gasoline are used to drive each mile.

Choice B is incorrect and represents the number of gallons of gasoline in a full gas tank.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Medium

Question ID da602115

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: da602115

SAT4 M 1.9

If $|4x - 4| = 112$, what is the positive value of $x - 1$?

ID: da602115 Answer

Correct Answer: 28

Rationale

The correct answer is **28**. The given absolute value equation can be rewritten as two linear equations: $4x - 4 = 112$ and $-(4x - 4) = 112$, or $4x - 4 = -112$. Adding **4** to both sides of the equation $4x - 4 = 112$ results in $4x = 116$. Dividing both sides of this equation by **4** results in $x = 29$. Adding **4** to both sides of the equation $4x - 4 = -112$ results in $4x = -108$. Dividing both sides of this equation by **4** results in $x = -27$. Therefore, the two values of $x - 1$ are $29 - 1$, or **28**, and $-27 - 1$, or -28 . Thus, the positive value of $x - 1$ is **28**.

Alternate approach: The given equation can be rewritten as $|4(x - 1)| = 112$, which is equivalent to $4|x - 1| = 112$. Dividing both sides of this equation by **4** yields $|x - 1| = 28$. This equation can be rewritten as two linear equations: $x - 1 = 28$ and $-(x - 1) = 28$, or $x - 1 = -28$. Therefore, the positive value of $x - 1$ is **28**.

Question Difficulty: Medium

Question ID 8f65cddc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 8f65cddc

SAT4 M 1.10

$$\frac{1}{7b} = \frac{11x}{y}$$

The given equation relates the positive numbers b , x , and y . Which equation correctly expresses x in terms of b and y ?

- A. $x = \frac{7by}{11}$
- B. $x = y - 77b$
- C. $x = \frac{y}{77b}$
- D. $x = 77by$

ID: 8f65cddc Answer

Correct Answer: C

Rationale

Choice C is correct. Multiplying each side of the given equation by y yields the equivalent equation $\frac{y}{7b} = 11x$. Dividing each side of this equation by 11 yields $\frac{y}{77b} = x$, or $x = \frac{y}{77b}$.

Choice A is incorrect. This equation is not equivalent to the given equation.

Choice B is incorrect. This equation is not equivalent to the given equation.

Choice D is incorrect. This equation is not equivalent to the given equation.

Question Difficulty: Medium

Question ID f7e39fe9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: f7e39fe9

SAT4 M 1.11

x	10	15	20	25
$f(x)$	82	137	192	247

The table shows four values of x and their corresponding values of $f(x)$. There is a linear relationship between x and $f(x)$ that is defined by the equation $f(x) = mx - 28$, where m is a constant. What is the value of m ?

ID: f7e39fe9 Answer

Correct Answer: 11

Rationale

The correct answer is 11. It's given that $f(x)$ is defined by the equation $f(x) = mx - 28$, where m is a constant. It's also given in the table that when $x = 10$, $f(x) = 82$. Substituting 10 for x and 82 for $f(x)$ in the equation $f(x) = mx - 28$ yields, $82 = m(10) - 28$. Adding 28 to both sides of this equation yields $110 = 10m$. Dividing both sides of this equation by 10 yields $11 = m$. Therefore, the value of m is 11.

Question Difficulty: Medium

Question ID 1dd13816

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	

ID: 1dd13816

SAT4 M 1.12

$$(5x^3 - 3) - (-4x^3 + 8)$$

The given expression is equivalent to $bx^3 - 11$, where b is a constant. What is the value of b ?

ID: 1dd13816 Answer

Correct Answer: 9

Rationale

The correct answer is **9**. The given expression can be rewritten as $(5x^3 - 3) + (-1)(-4x^3 + 8)$. By applying the distributive property, this expression can be rewritten as $5x^3 - 3 + 4x^3 + (-8)$, which is equivalent to $(5x^3 + 4x^3) + (-3 + (-8))$. Adding like terms in this expression yields $9x^3 - 11$. Since it's given that $(5x^3 - 3) - (-4x^3 + 8)$ is equivalent to $bx^3 - 11$, it follows that $9x^3 - 11$ is equivalent to $bx^3 - 11$. Therefore, the coefficients of x^3 in these two expressions must be equivalent, and the value of b must be **9**.

Question Difficulty: Medium

Question ID f2bbd43d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: f2bbd43d

SAT4 M 1.13

$$\begin{aligned} y &> 14 \\ 4x + y &< 18 \end{aligned}$$

The point $(x, 53)$ is a solution to the system of inequalities in the xy -plane. Which of the following could be the value of x ?

- A. -9
- B. -5
- C. 5
- D. 9

ID: f2bbd43d Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the point $(x, 53)$ is a solution to the given system of inequalities in the xy -plane. This means that the coordinates of the point, when substituted for the variables x and y , make both of the inequalities in the system true. Substituting 53 for y in the inequality $y > 14$ yields $53 > 14$, which is true. Substituting 53 for y in the inequality $4x + y < 18$ yields $4x + 53 < 18$. Subtracting 53 from both sides of this inequality yields $4x < -35$. Dividing both sides of this inequality by 4 yields $x < -8.75$. Therefore, x must be a value less than -8.75 . Of the given choices, only -9 is less than -8.75 .

Choice B is incorrect. Substituting -5 for x and 53 for y in the inequality $4x + y < 18$ yields $4(-5) + 53 < 18$, or $33 < 18$, which is not true.

Choice C is incorrect. Substituting 5 for x and 53 for y in the inequality $4x + y < 18$ yields $4(5) + 53 < 18$, or $73 < 18$, which is not true.

Choice D is incorrect. Substituting 9 for x and 53 for y in the inequality $4x + y < 18$ yields $4(9) + 53 < 18$, or $89 < 18$, which is not true.

Question Difficulty: Medium

Question ID 7e5a3640

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 7e5a3640

SAT4 M 1.14

Bacteria are growing in a liquid growth medium. There were **300,000** cells per milliliter during an initial observation. The number of cells per milliliter doubles every **3** hours. How many cells per milliliter will there be **15** hours after the initial observation?

- A. **1,500,000**
- B. **2,400,000**
- C. **4,500,000**
- D. **9,600,000**

ID: 7e5a3640 Answer

Correct Answer: D

Rationale

Choice D is correct. Let y represent the number of cells per milliliter x hours after the initial observation. Since the number of cells per milliliter doubles every **3** hours, the relationship between x and y can be represented by an exponential equation of the form $y = a(b)^{\frac{x}{k}}$, where a is the number of cells per milliliter during the initial observation and the number of cells per milliliter increases by a factor of b every k hours. It's given that there were **300,000** cells per milliliter during the initial observation. Therefore, $a = 300,000$. It's also given that the number of cells per milliliter doubles, or increases by a factor of **2**, every **3** hours. Therefore, $b = 2$ and $k = 3$. Substituting **300,000** for a , **2** for b , and **3** for k in the equation $y = a(b)^{\frac{x}{k}}$ yields $y = 300,000(2)^{\frac{x}{3}}$. The number of cells per milliliter there will be **15** hours after the initial observation is the value of y in this equation when $x = 15$. Substituting **15** for x in the equation $y = 300,000(2)^{\frac{x}{3}}$ yields $y = 300,000(2)^{\frac{15}{3}}$, or $y = 300,000(2)^5$. This is equivalent to $y = 300,000(32)$, or $y = 9,600,000$. Therefore, **15** hours after the initial observation, there will be **9,600,000** cells per milliliter.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

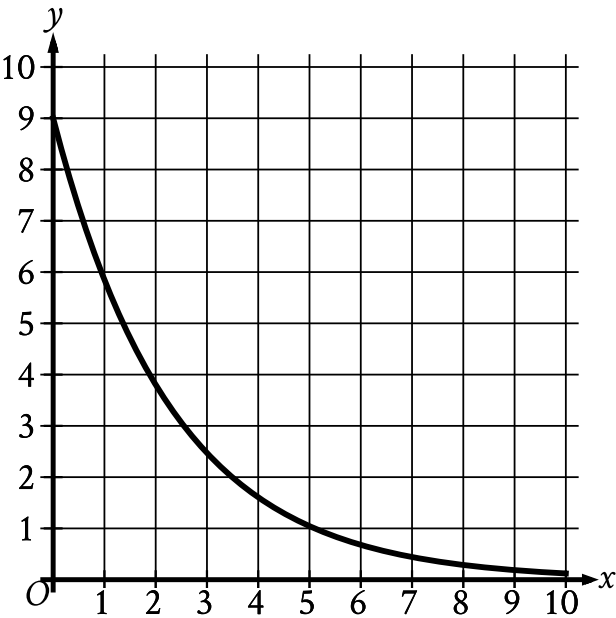
Question Difficulty: Medium

Question ID db888cd6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: db888cd6

SAT4 M 1.15



The graph gives the estimated number of catalogs y , in thousands, a company sent to its customers at the end of each year, where x represents the number of years since the end of **1992**, where $0 \leq x \leq 10$. Which statement is the best interpretation of the y -intercept in this context?

- A. The estimated total number of catalogs the company sent to its customers during the first **10** years was **9,000**.
- B. The estimated total number of catalogs the company sent to its customers from the end of **1992** to the end of **2002** was **90**.
- C. The estimated number of catalogs the company sent to its customers at the end of **1992** was **9**.
- D. The estimated number of catalogs the company sent to its customers at the end of **1992** was **9,000**.

ID: db888cd6 Answer

Correct Answer: D

Rationale

Choice D is correct. The y -intercept of the graph is the point at which the graph crosses the y -axis, or the point for which the value of x is **0**. Therefore, the y -intercept of the given graph is the point $(0, 9)$. It's given that x represents the number of years since the end of **1992**. Therefore, $x = 0$ represents **0** years since the end of **1992**, which is the same as the end of **1992**. It's also given that y represents the estimated number of catalogs, in thousands, that the company sent to its customers at the end of the year. Therefore, $y = 9$ represents **9,000**.

catalogs. It follows that the y -intercept $(0, 9)$ means that the estimated number of catalogs the company sent to its customers at the end of **1992** was **9,000**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 4eaf0a3a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	

ID: 4eaf0a3a

SAT4 M 1.16

Which expression is equivalent to $\sqrt[7]{x^9y^9}$, where x and y are positive?

- A. $(xy)^{\frac{7}{9}}$
- B. $(xy)^{\frac{9}{7}}$
- C. $(xy)^{16}$
- D. $(xy)^{63}$

ID: 4eaf0a3a Answer

Correct Answer: B

Rationale

Choice B is correct. For positive values of a and b , $a^mb^m = (ab)^m$, $\sqrt[n]{a} = (a)^{\frac{1}{n}}$, and $(a^j)^k = a^{jk}$. Therefore, the given expression, $\sqrt[7]{x^9y^9}$, can be rewritten as $\sqrt[7]{(xy)^9}$. This expression is equivalent to $\left((xy)^9\right)^{\frac{1}{7}}$, which can be rewritten as $(xy)^{9\cdot\frac{1}{7}}$, or $(xy)^{\frac{9}{7}}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 121dc44f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: 121dc44f

SAT4 M 1.17

The population of City A increased by **7%** from **2015** to **2016**. If the **2016** population is k times the **2015** population, what is the value of k ?

- A. **0.07**
- B. **0.7**
- C. **1.07**
- D. **1.7**

ID: 121dc44f Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the population of City A increased by **7%** from **2015** to **2016**. Therefore, the population of City A in **2016** includes **100%** of the population of City A in **2015** plus an additional **7%** of the population of City A in **2015**. This means that the population of City A in **2016** is **107%** of the population in **2015**. Thus, the population of City A in **2016** is $\frac{107}{100}$, or **1.07**, times the **2015** population. Therefore, the value of k is **1.07**.

Choice A is incorrect. This would be the value of k if the population in **2016** was **7%** of the population in **2015**.

Choice B is incorrect. This would be the value of k if the population in **2016** was **70%** of the population in **2015**.

Choice D is incorrect. This would be the value of k if the population increased by **70%**, not **7%**, from **2015** to **2016**.

Question Difficulty: Medium

Question ID b3c7ca1d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: b3c7ca1d

SAT4 M 1.18

Which of the following systems of linear equations has no solution?

- A. $x = 3$
 $y = 5$
- B. $y = 6x + 6$
 $y = 5x + 6$
- C. $y = 16x + 3$
 $y = 16x + 19$
- D. $y = 5$
 $y = 5x + 5$

ID: b3c7ca1d Answer

Correct Answer: C

Rationale

Choice C is correct. A system of two linear equations in two variables, x and y , has no solution if the graphs of the lines represented by the equations in the xy -plane are distinct and parallel. The graphs of two lines in the xy -plane represented by equations in slope-intercept form, $y = mx + b$, where m and b are constants, are parallel if their slopes, m , are the same and are distinct if their y -coordinates of the y -intercepts, b , are different. In the equations $y = 16x + 3$ and $y = 16x + 19$, the values of m are each 16 , and the values of b are 3 and 19 , respectively. Since the slopes of these lines are the same, and the y -coordinates of the y -intercepts are different, it follows that the system of linear equations in choice C has no solution.

Choice A is incorrect. The lines represented by the equations in this system are a vertical line and a horizontal line. Therefore, this system has a solution, $(3, 5)$, rather than no solution.

Choice B is incorrect. The two lines represented by these equations have different slopes and the same y -coordinate of the y -intercept. Therefore, this system has a solution, $(0, 6)$, rather than no solution.

Choice D is incorrect. The two lines represented by these equations are a horizontal line and a line with a slope of 5 that have the same y -coordinate of the y -intercept. Therefore, this system has a solution, $(0, 5)$, rather than no solution.

Question Difficulty: Medium

Question ID 2c6f214f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 2c6f214f

SAT4 M 1.19

The first term of a sequence is **9**. Each term after the first is **4** times the preceding term. If w represents the n th term of the sequence, which equation gives w in terms of n ?

- A. $w = 4(9^n)$
- B. $w = 4(9^{n-1})$
- C. $w = 9(4^n)$
- D. $w = 9(4^{n-1})$

ID: 2c6f214f Answer

Correct Answer: D

Rationale

Choice D is correct. Since w represents the n th term of the sequence and **9** is the first term of the sequence, the value of w is **9** when the value of n is **1**. Since each term after the first is **4** times the preceding term, the value of w is $9(4)$ when the value of n is **2**. Therefore, the value of w is $9(4)(4)$, or $9(4)^2$, when the value of n is **3**. More generally, the value of w is $9(4^{n-1})$ for a given value of n . Therefore, the equation $w = 9(4^{n-1})$ gives w in terms of n .

Choice A is incorrect. This equation describes a sequence for which the first term is **36**, rather than **9**, and each term after the first is **9**, rather than **4**, times the preceding term.

Choice B is incorrect. This equation describes a sequence for which the first term is **4**, rather than **9**, and each term after the first is **9**, rather than **4**, times the preceding term.

Choice C is incorrect. This equation describes a sequence for which the first term is **36**, rather than **9**.

Question Difficulty: Hard

Question ID 8f0c82e2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: 8f0c82e2

SAT4 M 1.20

The minimum value of x is 12 less than 6 times another number n . Which inequality shows the possible values of x ?

- A. $x \leq 6n - 12$
- B. $x \geq 6n - 12$
- C. $x \leq 12 - 6n$
- D. $x \geq 12 - 6n$

ID: 8f0c82e2 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the minimum value of x is 12 less than 6 times another number n . Therefore, the possible values of x are all greater than or equal to the value of 12 less than 6 times n . The value of 6 times n is given by the expression $6n$. The value of 12 less than $6n$ is given by the expression $6n - 12$. Therefore, the possible values of x are all greater than or equal to $6n - 12$. This can be shown by the inequality $x \geq 6n - 12$.

Choice A is incorrect. This inequality shows the possible values of x if the maximum, not the minimum, value of x is 12 less than 6 times n .

Choice C is incorrect. This inequality shows the possible values of x if the maximum, not the minimum, value of x is 6 times n less than 12, not 12 less than 6 times n .

Choice D is incorrect. This inequality shows the possible values of x if the minimum value of x is 6 times n less than 12, not 12 less than 6 times n .

Question Difficulty: Medium

Question ID c9931030

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	■ ■ ■

ID: c9931030

SAT4 M 1.21

$$RS = 20$$

$$ST = 48$$

$$TR = 52$$

The side lengths of right triangle RST are given. Triangle RST is similar to triangle UVW , where S corresponds to V and T corresponds to W . What is the value of $\tan W$?

- A. $\frac{5}{13}$
- B. $\frac{5}{12}$
- C. $\frac{12}{13}$
- D. $\frac{12}{5}$

ID: c9931030 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that right triangle RST is similar to triangle UVW , where S corresponds to V and T corresponds to W . It's given that the side lengths of the right triangle RST are $RS = 20$, $ST = 48$, and $TR = 52$. Corresponding angles in similar triangles are equal. It follows that the measure of angle T is equal to the measure of angle W . The hypotenuse of a right triangle is the longest side. It follows that the hypotenuse of triangle RST is side TR . The hypotenuse of a right triangle is the side opposite the right angle. Therefore, angle S is a right angle. The adjacent side of an acute angle in a right triangle is the side closest to the angle that is not the hypotenuse. It follows that the adjacent side of angle T is side ST . The opposite side of an acute angle in a right triangle is the side across from the acute angle. It follows that the opposite side of angle T is side RS . The tangent of an acute angle in a right triangle is the ratio of the length of the opposite side to the length of the adjacent side. Therefore, $\tan T = \frac{RS}{ST}$. Substituting 20 for RS and 48 for ST in this equation yields $\tan T = \frac{20}{48}$, or $\tan T = \frac{5}{12}$. The tangents of two acute angles with equal measures are equal. Since the measure of angle T is equal to the measure of angle W , it follows that $\tan T = \tan W$. Substituting $\frac{5}{12}$ for $\tan T$ in this equation yields $\frac{5}{12} = \tan W$. Therefore, the value of $\tan W$ is $\frac{5}{12}$.

Choice A is incorrect. This is the value of $\sin W$.

Choice C is incorrect. This is the value of $\cos W$.

Choice D is incorrect. This is the value of $\frac{1}{\tan W}$.

Question Difficulty: Hard

Question ID cc3e9528

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	■ ■ ■

ID: cc3e9528

SAT4 M 1.22

The graph of $9x - 10y = 19$ is translated down 4 units in the xy -plane. What is the x -coordinate of the x -intercept of the resulting graph?

ID: cc3e9528 Answer

Correct Answer: 59/9, 6.555, 6.556

Rationale

The correct answer is $\frac{59}{9}$. When the graph of an equation in the form $Ax + By = C$, where A , B , and C are constants, is translated down k units in the xy -plane, the resulting graph can be represented by the equation $Ax + B(y + k) = C$. It's given that the graph of $9x - 10y = 19$ is translated down 4 units in the xy -plane. Therefore, the resulting graph can be represented by the equation $9x - 10(y + 4) = 19$, or $9x - 10y - 40 = 19$. Adding 40 to both sides of this equation yields $9x - 10y = 59$. The x -coordinate of the x -intercept of the graph of an equation in the xy -plane is the value of x in the equation when $y = 0$. Substituting 0 for y in the equation $9x - 10y = 59$ yields $9x - 10(0) = 59$, or $9x = 59$. Dividing both sides of this equation by 9 yields $x = \frac{59}{9}$. Therefore, the x -coordinate of the x -intercept of the resulting graph is $\frac{59}{9}$. Note that 59/9, 6.555, and 6.556 are examples of ways to enter a correct answer.

Question Difficulty: Hard